

A1 production kernel manifest – scoped T5 promotion enactment (v1.7)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

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1. Purpose and scope

This note records the scoped T5 promotion of the canonical N-001 production-kernel manifest. The certified object is only the pure-Brazovskii scalar slice encoded in [claims/A1-PRODUCTION-KERNEL-MANIFEST/canonical_n001_kernel.json](#). The full anisotropic, locked, shell-biased, condensate, Hessian, PDE, BCC-selection, and unrestricted operator-theorem claims remain outside scope.

2. Verified runtime configuration

The verifier reads the six physical fields, three tolerances, and the complete seven-key `scalar_slice` from the canonical JSON. It imports the original vendored N-001 solver sources and obtains

$$\text{bloch_matrix_linear}(|k|, \{r, Z, Y, q_0, \text{**scalar_slice}\}) = r_{\text{zero}} + Z|k|^2 + Y|k|^4$$

with maximum residual 8.9×10^{-16} . The stored source hashes, the three independent consistency deltas, the positive shell mass, and the load-bearing scalar-slice settings are all checked.

3. Independent reproduction

The operator evidence package [claims/A1-PRODUCTION-KERNEL-MANIFEST/runs/promotion-evidence/20260716-operator-01/](#) records an independent REPRODUCTION-PASS. The reviewer checked the recorded command, `all_pass=true`, checker version, input hashes, constrained scope, no-overclaim boundary, and main-proof-line status. That review is the ratification that converts the earlier 14/14 gate pass into a scoped T5 result.

4. T5 bundle requirement

The June 2026 bundle directories are retained only as superseded pre-certification reproduction history. The current published package must be a claim-top-level T5-stamped bundle built from this note, the current verifier, the canonical JSON, the vendored original solver sources, and the current PASS output. The bundle does not enlarge the claim beyond this pinned scalar-slice scope.

5. Devil’s-advocate

(α) “*The checker chooses the scalar slice.*” DISMISSED – every setting is loaded from JSON and demonstrated load-bearing. (β) “*The vendored solver is not original.*” DISMISSED – all three source hashes are checked against provenance and config. (γ) “*Scalar-slice agreement proves the full operator.*” VALID with mitigation – the scope excludes the full operator, PDE behavior, BCC selection, and Hessian stability. (δ) “*T5 implies T6 or T7.*” UPHELD as a prohibited inference – this note provides no theorem-level discharge for the excluded components.

6. Result footer

Result ID:	A1-PRODUCTION-KERNEL-MANIFEST
Precise statement:	The canonical N-001 pure scalar slice passes the stored-field, runtime-symbol, positivity, and source-hash checks (14/14) and has independent operator reproduction.
Scope:	Pure-Brazovskii scalar slice only; full operator/PDE/BCC and Hessian claims excluded.
Dependencies:	A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Evidence grade:	ANALYTIC + EXECUTED; independent reproduction package.
Reproduction command:	python codes/foundations/a1_kernel_checks.py
Expected output:	A1 kernel checks v1.7.0: 14/14 PASS.
Falsification gate:	Config, scalar-slice, source-hash, delta, coefficient, or independent-review mismatch invalidates this scoped T5.
Tier before / after:	T1 / T5 (scoped pure-Brazovskii scalar-slice manifest).
No-overclaim statement:	No full-operator, PDE, BCC, Hessian, T6, or T7 claim.
Next required action:	Use only within pinned scalar-slice scope; separate theorem evidence is required for any future T6/T7 path.